

DATA ANALYTICS LEVEL III

Day 8: Importing & Structuring Data — CSV to SQLite, Data Types & NULLs

Student Activities

How to Use This Document

You are TechHub's analyst. HR just sent new_hires.csv — 12 people joining the company — and asked you to load it into the database. Files needed: Day08_practice.db and new_hires.csv from the LMS. For every query question, record BOTH the SQL and the answer.

Activity	What You Submit	Time
Activity 1: Build & Fill a Table	Your CREATE + INSERT script and row count	40 min
Activity 2: Import new_hires.csv + QA	Screenshot of import + 3 QA query results	50 min
Activity 3: NULL Hunt & First-Week Report	Queries + answers for all 10 questions	50 min
Learning Journal Entry #7	Your journal reflection	20 min

Activity 1 — Build & Fill a Table (CREATE + INSERT)

Goal: Prove you can define structure and load data by hand — the foundation of every import.

Connect SQLTools to Day08_practice.db (new connection named Day08). Then:

1. Create an interns table with these columns: intern_id (whole-number unique ID), full_name (text, REQUIRED), school (text, optional), start_date (text date). Choose the types and rules yourself — that IS the exercise.
2. Insert these three interns in ONE statement: Paula Vergara (PUP, 2024-06-10), Rico Dizon (UST, 2024-06-10), and Sam Aguilar — whose school is UNKNOWN (store a real NULL, not the word 'unknown').
3. Verify: how many rows? (COUNT). Which interns have a known school? (IS NOT NULL).
4. Prove your rules work: try inserting a 4th intern with NO name. Write down the exact error message you get.

Submit: Day08_Activity1_[YourName].sql (your statements + the error message as a comment)

Activity 2 — Import new_hires.csv (Guided) + the QA Ritual

Goal: Load HR's real file with DB Browser for SQLite and verify it like a professional.

Step 1 — Inspect before you import

5. Open new_hires.csv in Notepad or VS Code (right-click → Open With). How many data rows? Which rows have blanks, and in which columns? Write them down — this is your prediction for the NULL hunt.

Step 2 — Import

6. Open DB Browser for SQLite → File → Open Database → Day08_practice.db
7. File → Import → Table from CSV file... → new_hires.csv → table name: new_hires
8. TICK “Column names in first line” → check the preview → OK
9. Click “Write Changes”, then File → Close Database (so SQLTools can query it)

Step 3 — The QA ritual (in SQLTools)

10. Row count: `SELECT COUNT(*) FROM new_hires;` — does it match the file?
11. Eyeball: `SELECT * FROM new_hires LIMIT 5;` — columns aligned, values sane?
12. Sanity: `SELECT MIN(salary), MAX(salary) FROM new_hires;` — plausible pay range?

Submit: Day08_Activity2_[YourName].docx — import screenshot + the three QA results

Activity 3 — NULL Hunt & the First-Week Report

Goal: Find every hole in the imported data, then answer HR's real questions with Day 7 filters.

Part A — The NULL Hunt

13. Who has NO department assigned yet? (Show first_name, last_name. Remember: IS NULL.)
14. Whose salary is still missing? (One person — who?)
15. Run the three counts and explain the differences in one sentence: `COUNT(*)`, `COUNT(salary)`, `COUNT(department)`.
16. The trap, on purpose: run `WHERE department = NULL`. How many rows? Why? (One sentence.)
17. Run `SELECT first_name, salary FROM new_hires ORDER BY salary ASC;` — who appears FIRST, and what does that tell you about how SQLite sorts NULL?

Part B — HR's First-Week Questions (Day 7 skills on imported data)

18. Which new hires are based in Manila or Makati? (`IN` — show first_name, city.) How many?
19. How many were hired in 2024? (`LIKE` on hire_date.)

20. Who earns between ₱25,000 and ₱30,000? (BETWEEN — show first_name, salary.)
21. Top 3 paid new hires? (ORDER BY + LIMIT.)
22. What cities do the new hires live in — no repeats? (DISTINCT.) Compare with the old employees table: which city is NEW to the company?

Challenge

23. HR asks: “One list please — every new hire with ANY missing information (department or salary), so we can chase the paperwork.” Write ONE query. (Hint: two IS NULLs and an OR.)

Submit: Day08_Activity3_[YourName].docx (or .sql with comments)

Learning Journal Entry #7

Goal: Reflect on structure, imports, and missing data.

1. **In your own words:** what is NULL, and why is WHERE column = NULL the wrong way to find it?

2. List the three steps of the import QA ritual, and what each one catches.

3. Why does COUNT(*) say 12 but COUNT(salary) say 11 on the same table?

4. My confidence with importing data today (1 = lost, 5 = very confident), and why:

Submit as: Day08_Journal7_[YourName].docx

Submission Checklist

- [] **Activity 1:** interns table script + the NOT NULL error message
- [] **Activity 2:** import screenshot + 3 QA results

- ☐ **Activity 3:** all 10 NULL-hunt and report answers + the challenge query
- ☐ Learning Journal Entry #7